and act as a potential resource for combating caner.

6.6.3 Applications of NGS technologies

Apart from obtaining the sequence of DNA or genomes, the current sequencing strategies can be tailored for different applications. A few important applications are discussed below.

Transcriptomic Profiling (RNA-Seq)

In this method the RNA content in a cell can be isolated and sequenced in a quantitative fashion. As a result this method can profile the genes that are expressed in a cell in a given condition.

Polymorphism and Variation Discovery

This experiment identifies the mutations or variations in a genome sequence and associates them with phenotypes such as diseases.

Protein-DNA Interaction Analysis (ChIP-Seq)

This method is also known as Chromatin Immuno Precipitation (ChIP), which precipitates the protein (histone) DNA complexes alone, by means of specific antibodies and then that region of DNA alone is sequenced. This method helps us to identify the genome regions which can interact with proteins like histones.

Metagenomics

Metagenomics can be defined as the study of genetic material recovered directly from environmental samples. For example, the study of the microbiome in the gut of an individual. These samples contain extremely large but unknown numbers of unculturable microbial species. So such samples are directly subjected to DNA isolation and sequencing (without culturing) and considered as the genome of an entire community.

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